



Sanat Kumar Nayak

23+ Years of Experience in Engineering, Product Development, execution, Organization Strategy, Project Management, Operations, Maintenance & Business Development in Railway & Off-Highway Segment.

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Professional Summary & Work Experience

SOTECH India Pvt. Ltd Mar'24 to Present

Bid Engineering Leader (Feb'26 Onwards)

Key Responsibility

- Assigned a new responsibility from Top Leadership to Lead the Bid Engineering activities for upcoming opportunities in India & Globally.
- Responsibility includes Market Analysis, Product Strategy, Potential Customer Interactions & leading the Bid Technical Activities for all future opportunities in Mainline, Metros & other segments.
- Define a clear Bid Roadmap & mapping future opportunity with existing products with-in TMH Portfolio are the key responsibilities
- Participating with strategical discussion with potential JV's & Partnerships.
- Supporting commercial teams in preparation of additional configurations for improving the exportability opportunities as well overall product value propositions for specific segments of the market.

Head of Engineering Vande Bharath Sleeper (Mar'24 to Feb'26)

About KINET: Russian-Indian consortium (TMH-RVNL) signed a contract with Indian Railway (6.5 Bn\$) for the supply of 120 (16 cars) electric passenger trains with sleeping coaches for 180 kmph Design Speed to Indian Railways. According to the contract, production is to be established in Latur (Maharashtra) after the modernization of the Government Manufacturing Units.

Key Responsibility & accomplishment

- Head of Engineering of India's first Mainline Sleeper EMU for 180 Kmph Design Speed based on the **Vande Bharath Platform**
- Total Contract Value of 6.5 Bn \$ for Supply & Maintenance of 120 Trains of 16 Coaches for a Period of 35 Yrs.
- Total Engineering Budget of ~58m€ for Design & Development, Testing & Commissioning including start of Service Trials with Passenger Operations
- Direct reporting of 24 Highly Skilled Technical Resources, Indirect Reporting of 150+ Engineers across 5 Main Engineering Consultancies. 3 out of 5 are based outside India.
- Main decision maker for Product Definition & development, Industrialization strategy & Customer Management for design approvals, Technical Contract Management etc.
- Member of the Project Core Team to define the Project Planning, schedule & Risk Management, Configuration & Baseline Management.
- Build a Team with Highly Motivated Individuals from Scratch, defined Most of the Product development process including Roadmap for future projects

- In less than 2 Yrs a New Product has been defined which will replace existing Rajdhani Trains with best-in-class Modern Amenities, Passenger Comfort & class leading Performance comparable to top class EMUs globally.
- Opened a New Development Center in Hyderabad & vision to grow the Team for upcoming Tenders for domestic & International Projects
- Received high-level appreciation for Managing a complex development program with a very tight schedule with Design to Cost, Perceived Quality & Overall Life Cycle Cost optimization of the Product as per Indian operational conditions.
- More than 95% Design Submission to Indian Railways & Independent Safety Assessor were secured & received approvals.
- Production ongoing in Latur with First Carbody Shell rolled out in Mar'26 & ready for Fitouts

ALSTOM Transportation India Pvt. Ltd (Ex-Bombardier Transportation) Feb'17 to Oct'23

Chief Engineer (Regional Rapid Transit System – RapidX - NCRTC) (Mar'20 onwards)

Accomplishment

- Project is **first in the segment in India for commuter trains for 180 kmph Design, 160 Kmph Operational speed.**
- As the Chief Engineer Lead the complete development, Validation, and Homologation activities
- 80% of the development was during the Covid Lockdown Period, which was extremely challenging for a new product, with cross-functional team across 4 Global sites.
- Product Technical Authority for all area Vehicle, Sub-Systems & components including, Requirements management, Risk management & technical contract management with Customer, Suppliers etc.
- Accountable of Engineering development, Product Cost, Quality (Gate Reviews) & On time performance (Design & First Prototype Production)
- Accountable for Design Approvals with Customer to secure Cash Milestones, Production ramp-up by resolving critical design issues identified in 1st Prototype development and testing
- Accountable for securing Product cost variation from bid to Project with clear roadmap of the Product Configuration considering Testing, validation & serial production aspect.
- Achieved key milestone for first Prototype in Apr'22, Static Validation Dec'22 & Critical Safety Dynamic Validation by Mar'23
- Safety and statutory Approvals from 2 Internationally accredited Independent Safety Assessment agencies (CERTIFIER & TUV) & Indian Notified Govt. Agencies like CMRS, CCRS etc.
- Lead the Fleet Performance team for defect analysis & filed Issues, coordinated with Engineering, supply chain & PI teams to provide permanent rectification of the defects to improve reliability & safety
- Received Awards & recognition for the Seamless development & On time delivery of the 1st Train in Apr'22 as per contractual Milestone

Product Family Chief Engineer ESCOM (Entry Segment Commuter) Dec'18 to Mar'20

Accomplishment

- Product coverage all projects for India & Emerging Markets in commuter segment
- Accountable for Product Technical Specification, Cost & Standardization to provide a clear direction & optimized solution to all projects.
- Strategic alignment with global Platform, Sales & Marketing, Region, Configuration & Planning to baseline Product Configurations i.e. Base, Variants & Options
- Part of Core team to develop Product book & defining a clear Product road map for Off-cycle development (pre-bid)
- Fully aligned to Bombardier business Transformation process to define a systematic & clean start to kick-off Off-cycle & On-Cycle development

Head of Vehicle Architecture (VSSE) , BT Hyderabad (Feb'17 to Nov'18)

Accomplishment

- As Site Head of Vehicle Architecture developed a core team for Various domain on Architects (Mechanical & Electrical), Performance & System Engineers.
- Supported both ESCOM & Movia Maxx Product family for Bids & Off-cycle development
- Part of leadership team under site Engg. Director to develop the site as per organization mandate.
- Successfully lead the Design Team as Lead Vehicle Architect for all cross functional scope & Interfaces between BT & SIEMENS for Kanacharapara Bid/Off-cycle development
- Lead the Technical Discussion with RDSO during the Pre-bid Stage with Queries, clarification to modify & change the Technical Requirements
- Worked as a deputy to Product Chief Engineer during Kanacharapara Bid

ALSTOM Transportation India (Aug'14 – Feb'17)

Train Design Engineer Bids & Tender (Sep'15 – Feb'17) Noida, Shanghai Line 5, Bangkok, HCMT Melbourne, MEGA, IR EMU, IR-Kanacharapara and MMRC Mumbai)

Accomplishments

- Part of the Core Team for feasibility analysis for developing SMART St-St (Stainless Steel Carbody, Modular Welded) Platform for upcoming Wide Body Metro projects for India & APAC Projects. The activity includes detail feasibility study of New Modular Platform with impact analysis for industrialization, Engineering. **Work closely with Sourcing to develop new suppliers, Supplier Qualification, make & buy strategy for Carbody Modules etc.**
- Developed a core team to work on Tenders & Pre-Engineering for APAC Projects.
- Part of Core Team for developing Variants (REFSOL: Indian based Reference Solution) based on Kochi & Lucknow Platform with Cost reduction (Design to cost), Standardization & Benchmarking against competitor product.
- Started a new Initiative for introducing VAVE (Value Engineering & Value Analysis) to Alstom Products line especially Kochi & Lucknow platform for Weight & Cost reduction, standardization.
- Part of the Core Team for bringing Central Engineering initiative for IBS (Integration break-down structure) strategy to Indian Projects as well as Piloting a new Engineering Estimation tool (Pleide) for all upcoming tenders

Responsibility

- TDE for all bids, that includes development of vehicle architecture i.e. Preliminary & final concept based on Gauging, Clearance with Compliance to technical requirements.
- Review & Finalize Vehicle Architecture Requirements Management to prepare General Vehicle concept, various Vehicle performance impacting architecture, Underframe & Roof architecture arrangement, Noise & Acoustic, Fire and Weight Management, vehicle dynamics, gauging etc.
- Review & finalize high level Product Concept for Electrical Integration, Pneumatic & Piping, Inter Car connection like Coupler & Gangway, Inter-CAR Harness (jumper), Inter-Unit architecture etc.
- For new Platform introduction, involve in the product development process with Platform Engineering like Carbody erection strategy, Crash requirements, emergency evacuation strategy for passenger safety.
- Review & Co-Ordinate with TSS, Interior, Design-Styling for formulation of the Product Styling & Design Concept. Involved in the decision-making process for Technical Delivery Specification for Center of Excellency Interiors, Design & Styling, Make or Buy strategy for CBS & Metallic Parts
- Authorize Technical Purchase Specification, EBoM & Connector files for Catalogue Items Commodities
- Support to Bid Management in Preparation of Win Go / No-Go Product Concept, Go Concept, Go-Specification (Vehicle Architecture & Integration Management)
- Perform detail GAP analysis from Reference Solution, implementation Reference Solution & Catalogue Products. Finalize Detail Clause by Clause, list of Installation with Carry over Analysis, Technical GAP, Risk, Complexity & Assumption etc.
- Prepare Product Break down structure for Tender Costing, Engineering Estimation, Delivery Planning with Co-ordination TD Domains

Lead Engineer Electrical Integration & Fittings – Kochi Metro (Aug'14 – Aug'15)

Accomplishments:

- Developed a new Team for Electrical Integration, Cabinet Design at Alstom Engineering Center Bangalore that includes hiring of Key Resources, define way of working etc.
- Lead & executed the first Engineering Metro Project from Alstom India Kochi Metro in a highly challenging timeline with completely new team.
- Cross Function with Project, Industrial, Production & other Cross Métier for inputs & deliverables, design strategy & deliverable as per Project plan
- Lead Design, Gate Reviews & responsible for final QCD of all Electrical Installations for Kochi Metro.

Larsen & Toubro (April'11- August '14)

Account Delivery Manager Railway Projects Technical Project Manager for all Mechanical Projects Additional responsibility of Project Management & Operations. Key stakeholder in the Strategy & growth of Railway Engineering in the Transportation Vertical

Accomplishments: Key Projects handled (Customer wise)

- **TEXMACO:** Design Optimization of new BCCW Cement 22.9 T axle load of Payload 68 T. The final design has been analyzed with FEA & design optimization recommendation has been provided to TEXMACO for RDSO approval.
- **WABTEC Passenger Transit:** S/W Development of brake control unit, Design & Development of Electric Coupler Test Bench Unit for Testing & Validation. Development Engineering Tools for Oil Free

Compressor Design Calculation etc. New Product Development of Air Tank (Pressure Vessel in Aluminum) , Indian supplier base development for Casting components etc.

- **THALES Transportation:** Site Technical Manager for Signaling Upgradation Brown filed project SMRT Singapore. The area of work includes Retrofit Design & investigation, dismantling process & Installation strategy for On-board ATP systems to KHI, KNS, SIEMENS Metro as well as work vehicles (Locomotives).
- **Ansaldo STS India:** Indian Mainline Signaling upgradation projects for North Central railway
- **IVECO, FIAT, Italy:** VAVE Specialist for Cost Reduction & Standardization program for EURO CARGO & STARLIS tractor trailer range. Achieved 20-41 Euro Per Door Saving as well as one common door platform for both the series with Annual Qty. more than 200 K
- **CNH, FIAT USA:** Started a new engagement with CNH site at Racine, Wisconsin USA for Value Engineering activity in the area for Parts rationalization, De-Complexity, Reverse Engineering for complete range of CNH agricultural products.

InfoTech Enterprises (May'05– March'11)

Project Leader (BOMBARDIER TRANSPORTATION Engineering Center India, Locomotive)

- Responsible & worked in the area of Locomotive Vehicle Layouts especially integration of HV, Auxiliary Supply & LV Panels in Design & Electrical –Mechanical integration.
- Design of HV Cubicle like Cubicle Layout, Arrangement of HV, LV components as per Function, Cabling, EMI/EMC & Insulation Design requirements
- Arrangement of Electrical Switchgears as per wiring, Ducting, Safety, Accessibility, Maintainability & Homologation point of View
- Experience in Selection of Modular Cubicles from Rittal, Hoffman etc.
- Sound knowledge & understanding in reading Electrical Schematics, Wiring diagrams & Wiring table etc.
- Experience in all kind of Mfg. Drawing & BT PDM BOM preparation for Carbody & Electrical Integration area i.e. Single Parts, Assembly, Welding Drawings as per EN 15085
- Experience in Cross functional co-ordination between design, Wiring & System Engineering teams
- Lead Engineering team from Hyderabad for Electromechanical integration activity on **TALGO 250, 350 & TRAXX platform locomotive**

Central Institute Tool Design, Hyderabad (April'04– April'05)

Engineer CAD & CAM

- Designer for in the In-house Production & Design Center Customers like DRDO, ARDE Pune, ISRO & Indian Private Sectors
- Instructor for CAD&CAM Software for M Tech, Post Graduate, B Tech students & corporate executives
- Manufacturing Programmer for In-House Production CNC
- Program Co-Ordinator for Key Training Courses

- Actively Engaged In-House Campus Interview Co-Ordination for MTech & Post Graduate Training Programs with Various Indian Corporates

Central Tool Room & Training Center, Bhubaneswar (Sep'02– April'04)

Instructor CAD CAM Tools

- Worked as Instructor for CAD&CAM Software Courses for B Tech & Post Graduate Students
- Also actively involved in the in-house Production & Design Center for projects like single components manufacturing, Tool Design & production Plastic moulds, Press Tools. NC Program generation for Customers like ISRO, ADA, NTPC, HAL, L&T etc.

Overseas Assignments

Bombardier Transportation, Zurich Switzerland

- Involved in Production documentation for Talgo & Traxx series locomotive
- Design arrangement in the Roof structure with critical verification of Air gap, insulation distance etc. Traction Motor cable routing, ensuring with the right slack, bending, protection from moving & flying objects from the track
- Designs for different antenna mountings in the under frame (European rails Traffic management system)
- Production drawings for cables routing in the Machine Room side wall & driver's cab. Final layout drawings for cubicle arrangements, earthing & Interface points to the Locomotive

Bombardier Transportation, Mannheim Germany

- Worked for retrofit projects from BR 185.0 platform in different structure like Carbody E/M interfaces, Driver's Cabinet, Roof & Side wall.
- Low voltage, high voltage & electronic cabinet
- Interface to the Off shore project team
- Training at Production site Kassel for Pre & final assembly of TALGO & Traxx platform locomotives.
- Training at Production site Mannheim for Main Converter assembly

CNH Racine, Wisconsin, USA

- Value Engineering, De Complexity, Parts Rationalization
- Reverse Engineering & Competitive Benchmarking

THALES, SMRT Depot, Singapore

- Site Program Manager for all Engineering Activity from Engineering Design Installation, Testing Commissioning
- Re-signaling& Retrofit for CBTC Up gradation Program in Rolling stock, Way-Side & On-Board
- Participate in all PDR, FDR & FAI with End Customer Singapore Metro

ALSTOM, VPF (Valenciennes) France

- On-Job Training for Alstom way of working in the Electrical Integration area
- Worked for Ottawa LRV

ALSTOM, STO (Paris) France

- Product Definition workshop with Platform & Expert Domain for HCMT Melbourne

- Product definition for SMART St-St Platform for all Stainless-Steel Metro Markets & Opportunity

Bombardier Transportation, Hennigsdorf, Germany

- Product Definition workshop & alignment with Mainline Product Family for convergence
- Product definition of S-Bahn Munich bid & Learning for Off-cycle development from SDDD Product

Professional Trainings & Workshop

- ISO 9000 & TQM program at CTTC, Bhubaneswar
- PDM key user training
- Undergoing German language training Level- A1
- C++ & SQL Server Training
- Certificate course on Advance Welding & Metrology (Advance Training Institute, Hyderabad)
- Finance for Non-Financial Managers in L&T Leadership Development Academy, Lonavala
- First Time Manager in L&T Technology Service
- Key Account Management L&T Technology Service
- ICONE Functional tolerance methods & applications
- Certification on Advance Welding & Metrology from Advance Training Institute Hyderabad -2010

Educational Summary

- Bachelor of Engineering (Mechanical) from Kalinga Institute of Technology & Science, Utkal University Orissa - 2002
- Master Certification in CAD&CAM from Central Tool Room & Training Center, Bhubaneswar - 2002

"All the Details furnished in this document are true and correct to the best of my knowledge"

*Thank You
Sanat Kumar Nayak*